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CONTAINER TREATMENT SYSTEM

Abstract

A container treatment system for treating containers, comprising a first treatment block with a first single-line container exit point, further comprising a downstream second treatment block having a wet region and/or temperature control region with a mass conveyor for containers, a first single-line container feed point upstream of the mass conveyor and a second single-line container exit point downstream of the mass conveyor, and comprising a third treatment block that is located downstream of the second treatment block and has a second single-line container feed point, wherein the first single-line container exit point of the first treatment block is connected to the first single-line container feed point of the second treatment block via a first single-line conveying device, and the second single-line container exit point of the second treatment block is connected to the second single-line container feed point of the third treatment block via a second single-line conveying device.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application is a U.S. National Phase of International Application No. PCT/EP2023/057927 entitled “CONTAINER TREATMENT SYSTEM,” and filed on Mar. 28, 2023. International Application No. PCT/EP2023/057927 claims priority to German Patent Application No. 10 2022 108 754.9 filed on Apr. 11, 2022. The entire contents of each of the above-listed applications are hereby incorporated by reference for all purposes.

TECHNICAL FIELD

[0002] The present invention relates to a container treatment system according to independent claim **1** and an assortment of container treatment systems according to claim **12** as well as a method for treating containers, such as bottles, according to claim **14**.

BACKGROUND AND SUMMARY

[0003] Container treatment systems, in particular block-based container treatment systems, are already known from the prior art. In this case, various container treatment machines, for example a filler and a capper and a downstream pasteurizer as well as a downstream decoration device such as a printing machine or a labeling machine, are usually operated at least partially synchronized so that the containers are continuously conveyed through the system and treated in the process. Conveying devices are arranged between the individual container treatment machines of a container treatment system. In order to catch any failures of individual container handling machines without having to stop the container handling machines arranged upstream or downstream or without adversely affecting their operation, extensive buffer systems are provided in the conveying devices arranged between the machines. These can accommodate a large number of containers and therefore at least partially maintain the operation of the container treatment system, for example in the event of a failure of a downstream container treatment machine or in the event of a failure of an upstream container treatment machine.

[0004] For this purpose, the buffer systems must be large enough to provide sufficient buffer capacity at a typical throughput rate of several thousand or several tens of thousands of containers per hour in order to avoid the adverse effects of a production interruption in a system described above.

[0005] This entails considerable space requirements as well as restrictions regarding the positioning of the container handling machines relative to each other, since the buffer systems have to be provided in between them.

Object

[0006] On the basis of the known prior art, the technical object to be achieved is to essentially maintain the operation of a container treatment system with a plurality of container treatment machines even in the event of a failure of the individual container treatment machines, while keeping space requirements as low as possible.

Solution

[0007] This object is achieved according to the invention by the container treatment system according to independent claim **1** as well as the assortment of container treatment systems according to claim **12** and the method for treating containers according to claim **14**. Advantageous

developments of the invention are disclosed in the dependent claims.

[0008] The container treatment system according to the invention for treating containers such as bottles comprises a first treatment block with a first single-line container exit point, a second treatment block arranged downstream comprising a wet region and/or temperature control region with a mass conveyor for containers or a buffer system for containers, a first single-line container feed point upstream from the mass conveyor or the buffer system, and a second single-line container exit point downstream from the mass conveyor or the buffer system, and a third treatment block arranged downstream from the second treatment block with a second single-line container feed point, wherein the single-line container exit point of the first treatment block is connected to the single-line container feed point of the second treatment block by a first single-line conveying device, and the single-line container exit point of the second treatment block is connected to the single-line container feed point of the third treatment block by a second single-line conveying device.

[0009] The invention is not limited to exactly one single-line container exit point or container feed point in the respective treatment blocks. A plurality (for example two, three or more) of single-line container exit points and/or single-line container feed points can thus be provided for respective treatment blocks, which can be arranged horizontally and/or vertically offset from one another.

[0010] Within the scope of the invention, a treatment block is understood to mean an arrangement of at least one, but possibly also a plurality of, container treatment machines which carry out a treatment of containers jointly or in isolation and are not separated from one another by the first or second single-line conveying device. In particular, within a treatment block, transportation can be effected by means of other conveying devices, in particular transport stars. Preferably, the throughput rate of the container treatment machines within a treatment block, is the same and/or the operation of the container treatment machines of a treatment block is synchronized so that the same number of containers are treated per unit of time by each container treatment machine of a treatment block.

[0011] The mass conveyor of the wet region and/or temperature control region is to be understood as a conveying device in which the containers are transported essentially in an unordered manner. According to the invention, this mass conveyor extends through the wet and/or temperature control region or is a part thereof.

[0012] In contrast to the mass conveyor, a single-line conveying device is understood to be a conveying device (such as a conveyor belt or air conveyor) in which containers can only be transported in exactly one line, i.e. one container after the other.

[0013] A wet region and/or temperature control region is also understood to mean any device that either heats and/or cools the containers and/or applies a liquid thereto. These include in particular pasteurizers and washing machines for containers.

[0014] In contrast, the buffer system is to be understood as being outside a treatment facility and in particular outside a wet region and/or temperature control region. In this sense, the buffer system only achieves the objective of buffering or storing containers after and/or before they are subjected to further container treatment.

[0015] According to the invention, the mass conveyor or a part of the mass conveyor is advantageously used as a buffer region, since a large number of containers can be transported in this mass conveyor. This means that additional buffer apparatuses downstream or upstream from the second treatment block can be eschewed, which reduces the space requirements for the conveying devices between the treatment blocks and therefore the container treatment system as a whole. In particular, this not only allows space to be saved but also advantageously allows a flexible arrangement of the treatment blocks relative to each other by appropriate settings or a design of the first and second single-line conveying devices. Single-line transport systems are more flexible in terms of their spatial arrangement and space requirements than buffer regions which must meet a specified buffer capacity (number of containers that can be accommodated).

[0016] The buffer system provided for this as an alternative can, as a treatment block or as part of the treatment block, advantageously implement a storage of containers while these can still be fed to this buffer system via the single-line conveying devices or removed therefrom by means of single-line conveying devices. With this embodiment, space is also saved, and a flexible arrangement of the treatment blocks is made possible.

[0017] This means that changing space requirements, such as for different customers, can be taken into account, although the design of the individual treatment blocks can always remain the same.

[0018] In one embodiment, the wet region and/or temperature control region is designed as a pasteurizer, and/or the second treatment block comprises a blower for blowing off liquid on the surface of the containers, or the buffer system comprises a line buffer comprising at least two single-line buffer tracks, or a mass buffer for random buffering of containers.

[0019] Since in a pasteurizer the containers are usually transported in a mass flow, this is particularly suitable for achieving the necessary buffering effect. It can further be provided that a region is arranged at the inlet and/or outlet of the pasteurizer in which containers can be buffered in an unordered manner after they have left the pasteurizer or before they are fed thereto. This can prevent containers from remaining in the pasteurizer for too long, even given a large buffering requirement. Line buffers and mass buffers can advantageously provide large buffer capacities.

[0020] Furthermore, it can be provided that the first single-line container exit point, the first single-line conveying device and the first single-line container feed point are designed for the single-line transportation of containers in at least two vertical transportation routes; and/or that the second single-line container exit point, the second single-line conveying device and the second single-line container feed point are designed for the single-line transportation of containers on at least two vertical transportation routes.

[0021] This design increases the throughput rate of the container treatment system and in particular the transport capacity of containers between the individual treatment blocks. It can be provided that a device for overcoming height is assigned to the treatment blocks at the inlet and/or outlet so that the containers transported on different vertical transportation routes are brought back to a height level before they are fed to a treatment machine of the respective treatment block or are distributed over two different height levels in order to be fed to the respective transportation routes after leaving the treatment block at the outlet. In particular, the second treatment block can comprise a mass conveyor which allows a treatment of containers at two height levels corresponding to the vertical transportation routes so that a device for overcoming the height is not required in this region.

[0022] In one embodiment, it is provided that the first treatment block comprises a filler, and/or a capper, and/or an inspection device for inspecting the containers.

[0023] These devices can advantageously be arranged outside a treatment block that includes a wet region and can therefore continue to operate even if, for example, the third treatment block fails, since buffering of the containers is realized in between them.

[0024] Furthermore, it can be provided that the third treatment block downstream from the second container feed point comprises a lane divider for dividing the containers from the second single-line conveying device into a plurality of lanes, and/or wherein the third treatment block comprises an inspection device for inspecting the containers, and/or wherein the third treatment block comprises a labeling machine and/or a direct printing machine.

[0025] By division of the lanes, the throughput rate of the third treatment block can be increased. Since the further treatment of the containers following a pasteurization is usually not time-critical (i.e. a certain standing time of the containers before this treatment is also possible without adversely affecting the containers), labeling machines or direct printing machines that are designed to decorate the containers can advantageously be arranged in this treatment block.

[0026] It can further be provided that the second treatment block and/or the third treatment block comprises a coding device for applying a code to containers.

[0027] By providing appropriate coding devices, it is possible to track the containers through the container treatment system, even if the containers are transported without any order in the mass conveyor and therefore in the region of a buffer for the container treatment system. If a coding device is arranged in the first as well as in the third treatment blocks, this allows precise tracking of where each container was treated in the first and third treatment blocks, despite the possible mixing of the containers in the second treatment block.

[0028] In one embodiment, no treatment of containers takes place in the region of the first conveying device and/or in the region of the second conveying device. The first and second conveying devices therefore only serve to transport the containers without any treatment of the containers taking place at the same time.

[0029] It can further be provided that upstream from the first treatment block, a fourth treatment block is arranged with a wet region and/or temperature control region, comprising a mass conveyor for containers, or with a buffer system for containers, wherein the fourth treatment block comprises a third single-line container exit point, and the first treatment block comprises a third container feed point, wherein the third single-line container exit point is connected to the third single-line container feed point by a third single-line conveying device.

[0030] The wet region and/or temperature control region with the mass conveyor can also serve as a buffer here to compensate for a failure of the downstream first treatment block.

[0031] In particular, the fourth treatment block can comprise a container washing machine if it is designed as comprising a wet region and/or temperature control region. Since the treatment of the containers in a container washing machine is not time-critical, a longer residence time and therefore a buffer effect of the container washing machine is possible without adversely affecting the containers in order to compensate for any failure of downstream container treatment machines.

[0032] It can further be provided that the third single-line container exit point, the third conveying device and the third single-line container feed point are designed for the single-line transport of containers on at least two vertical transportation routes. This allows the throughput rate of containers in the single-line transport systems to be increased.

[0033] In one embodiment, the fourth treatment block and/or the second treatment block comprise a buffer region for the random buffering of containers outside the wet region and/or temperature control region and downstream from the container feed point of the treatment block and upstream from the container exit point of the treatment block.

[0034] By providing external buffer regions, an excessively long treatment of the containers can be avoided.

[0035] According to the invention, an assortment of container treatment systems is also provided, wherein each of the container treatment systems comprises a container treatment system according to one of the preceding embodiments, and wherein for two different container treatment systems of the assortment of container treatment systems, the first single-line conveying device and/or the second single-line conveying device and/or the third single-line conveying device have different forms.

[0036] Advantageously, the container treatment systems can only vary with regard to the design of the transport equipment. This makes it possible to configure different container treatment systems to spatial conditions without having to change the individual treatment blocks significantly.

[0037] In particular, it can accordingly be provided that, for two different container treatment systems in the assortment of container treatment systems, the treatment blocks are designed substantially the same.

[0038] With this design, a flexible adaptation of the container treatment system to different spatial conditions is also made possible without requiring a fundamentally new design of the entire system, including the treatment blocks.

[0039] It can also be provided that, for two different container treatment systems in the assortment of container treatment systems, the treatment blocks are designed substantially the same. The fact

that the treatment blocks are designed substantially the same means in particular that they differ, if at all, only with regard to the container feed point and/or the container exit point, for example the position or orientation of the container feed point and/or the container exit point. Since these elements can be designed as a modular part of a treatment block, even changing these elements does not call for a fundamentally new design of the treatment block itself, which increases flexibility in providing a container treatment system to different spatial conditions, but at the same time reduces the effort involved in designing the container treatment systems.

[0040] Furthermore, a method for treating containers such as bottles is provided, wherein the containers are treated in a container treatment system according to one of the preceding embodiments, and at least one treatment step of the containers is carried out in the first treatment block, the second treatment block and the third treatment block.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0041] FIG. 1 shows a schematic representation of a container treatment system according to one embodiment.

[0042] FIGS. 2a and b show schematic embodiments of container transportation in the conveying devices and the treatment blocks.

[0043] FIG. 3 shows an embodiment of a buffer system

[0044] FIGS. 4a to c show different implementations of container treatment systems of an assortment of container treatment systems.

DETAILED DESCRIPTION

[0045] FIG. 1 shows a schematic view of a container treatment system **100** for treating containers **180** such as bottles, cans or the like according to an embodiment. In the view shown in FIG. 1, the container treatment system comprises a first treatment block **101** and, downstream in the transportation direction of the containers **180** of the first treatment block, a second treatment block **102** and, downstream in the transportation direction of the containers **180** of the second treatment block **102**, a third treatment block **103**.

[0046] Each of these treatment blocks **101** to **103** comprises at least one container treatment machine which can carry out a treatment step on the containers. Treating containers includes any action carried out on the containers, in particular filling, closing or cleaning, as well as heating or cooling or applying a liquid, but also equipping the containers with decorative elements such as labels or printed images. Inspecting the containers can also be understood as treating the containers. Likewise, the application of a code or other identification marking, for example to track individual containers through the treatment system, as well as reading such a code can be understood as treatment of the containers. However, mere buffering can also be understood as container treatment for embodiments in which in particular the second treatment block **102** comprises a buffer system or is designed as such (i.e. does not comprise any further container treatment machine). In particular as part of the first and/or third treatment blocks **101** and **103**, a coding device for applying a code to the containers, and/or a reading device for reading such a code, and/or an inspection device for inspecting the containers can thus also be provided.

[0047] The first treatment block **101** can thus comprise at least one container treatment machine in the form of a carousel **113**, indicated here schematically, along the periphery of which a plurality of container receptacles and/or treatment stations for treating the containers are arranged (not shown in detail here). The treatment machine can be designed in particular as a filler or capper or a combined filler-capper treatment machine. The first treatment block **101** can also comprise container treatment machines arranged one after the other (in the transportation direction of the containers), wherein each of these machines can be designed as a carousel and/or as a linearly

operating machine. For example, a filler can be designed as a linearly operating machine and a subsequent capper can be implemented as a rotary machine.

[0048] In one embodiment, the second treatment block **102** comprises a wet region and/or temperature control region **123**. This is designed as comprising a mass conveyor so that the containers are transported in an unordered manner in the region of the wet region and/or temperature control region **123**. The mass conveyor can have a length of a plurality of meters (for example 5 m or **10** m or more) and can be designed as a conveyor belt or transport belt on which containers are guided through the wet region and/or temperature control region **123**. The conveyor belt can in particular have a width transverse to the transportation direction of the containers **180** in the region **123** which is a multiple of the diameter of a container, so that a plurality of containers can be transported next to one another and in an unordered manner.

[0049] Preferably, the wet region and/or temperature control region can be designed as a pasteurizer and realize heating and/or cooling of the containers **180** guided through the wet region and/or temperature control region **123** either by radiation and/or air or by exposure to a liquid medium (for example water).

[0050] The second treatment block **102** can further preferably comprise a blower downstream from the wet region and/or temperature control region, which can cause the containers to dry if the containers were exposed to a liquid in the wet region and/or temperature control region **123**.

[0051] Alternatively, as already described, the second treatment block can comprise a buffer system **123** instead of a wet region and/or temperature control region. This buffer system can be designed in particular as a line buffer or as a mass buffer. A line buffer is understood as a buffer system in which containers can be temporarily stored in several lines (also called lanes) in an ordered manner, at least relative to the respective line. In a line buffer, the movement of containers in at least one of the lines of the line buffer can come to a standstill, although containers are continuously being fed into the line buffer and/or containers are continuously being removed from the line buffer. Preferred embodiments of such line buffers or line buffer tables are known from WO 2019/011516 A1, the content of which is hereby incorporated in its entirety by way of reference. In this regard, more detailed embodiments will be described with reference to FIG. **3**.

[0052] A mass buffer can be designed analogously to the mass conveyor in the wet region and/or temperature control region, wherein the mass buffer, in contrast to the mass conveyor of the wet region and/or temperature control region, is not designed as part of another container treatment machine, but merely realizes the function of buffering containers. The mass buffer can be designed in particular as a conveyor belt for the unordered transport of containers. Preferred embodiments of a mass buffer are described in EP 3 929 114 A1. The content of EP 3 929 114 A1 is hereby incorporated in its entirety by way of reference. Particularly preferred embodiments of a mass buffer are shown in EP 3 929 114 A1 in FIGS. **1** to **9** and described in the associated description in paragraphs 59 to 167. The methods described in connection with these embodiments are also preferred as embodiments of a mass buffer and in particular its operation.

[0053] The third treatment block **103** can then comprise one or more decoration devices and/or inspection devices. These can also be designed, as shown here, as one or more carousels **133**, along the periphery of which container receptacles and/or container treatment stations are arranged in order to carry out the respective treatment step. For example, given a design of the treatment block **103** as a labeling machine comprising a carousel **133** with container receptacles, it can be provided to arrange fixed labeling units along its periphery in order to apply the labels to the containers transported in the carousel. Alternatively or additionally, a printing machine, in particular a direct printing machine, can also be provided as part of the third treatment block **103**. This can, for example, include co-rotating print heads (i.e. arranged on the carousel) or fixed print heads that apply printing ink to the containers.

[0054] Optionally, a fourth treatment block **104** can be arranged upstream from the first treatment block **101**, which can comprise a mass conveyor **143** as part of a wet region and/or temperature

control region **143**. As an alternative to a wet region and/or temperature control region, a buffer system can also be provided here, for example in the form of a line buffer or mass buffer. The fourth treatment block **104** can in particular comprise a container washing machine as a realization of the wet region and/or temperature control region **143**. Analogously to the second treatment block **102**, it can also be provided here that a blower is arranged downstream from the wet region and/or temperature control region **143**, which carries out drying of the containers.

[0055] According to the invention, single-line conveying devices for single-line transport of the containers between the individual treatment blocks are arranged between the respective treatment blocks. A first single-line conveying device **152** is thus arranged between the first treatment block **101** and the second treatment block **102**. A second single-line conveying device **153** is arranged between the second treatment block **102** and the third treatment block **103**, and a third single-line conveying device **151** can be arranged between the optionally provided fourth treatment block **104** and the first treatment block **101**.

[0056] The conveying devices **151**, **152** and **153** can in particular be designed as conveyor belts which transport containers one after the other in only one line (seen transversely to the transportation direction of the containers in the respective conveying device). Alternatively or additionally thereto, one or more of the conveying devices can also be designed as air conveyors in which the containers are arranged hanging on the support ring in a guide channel and are transported in the transportation direction by an air flow.

[0057] The first treatment block **101** further comprises a first container exit point **112** which transfers containers from the treatment block **101** to the first single-line conveying device **152**. Here, for example, a suitable transfer of the containers from a transportation by neck handling or from a transportation in corresponding container receptacles within the first treatment block **101** to the single-line first conveying device **151** can take place. Corresponding means are known from the prior art.

[0058] The second treatment block **102** comprises a first container feed point **121** via which the containers from the first single-line conveying device **152** are transferred to the second treatment block **102**. This first container feed point **121** can comprise one or more guide elements or gates in order to distribute the containers from the single-line transport to the mass conveyor. For example, guide gates can be used for this which fan out in the transportation direction of the containers so that an unordered mass transportation or a corresponding handover to the mass conveyor is ensured as far as possible across the entire width of the mass conveyor.

[0059] The second treatment block **102** further comprises, downstream from the wet region and/or temperature control region **123** or the buffer system, a second container exit point **122** which transfers the containers from the mass conveyor to a downstream second single-line conveying device **153**. For this purpose, the second container exit point **122** can be designed “inversely” to the first container feed point **121**, so that it separates into singles the containers from the initially unordered transportation in the mass conveyor **123** and causes a transfer to the second single-line conveying device. For this purpose, suitable guide gates can also be provided which accommodate the containers in a plurality of lines, which are then brought together to ensure transportation in just one line.

[0060] In one embodiment, it can be provided that the first container feed point **121** and/or the second container exit point **122** comprise buffer regions not shown in detail here, which can basically be designed as standing surfaces or conveyor belts and can be used as buffers if container treatment is interrupted downstream or upstream from the second treatment block (for example due to a malfunction). It is then possible to block the second container exit point and to buffer the incoming containers in the buffer region but also in the mass conveyor until they have reached their respective capacity limit. It can be provided that the buffer regions assigned to the first container feed point **121** and/or the second container exit point **122** can accommodate up to 10,000 or up to 20,000 containers or more.

[0061] The second single-line conveying device **153** then feeds the containers to the third treatment block **103**, wherein this comprises a second container feed point **131** at which the containers can be taken by the second single-line conveying device **152** and fed to a container treatment machine of the third treatment block **103**. Here, for example, the containers transported in the conveyor belt of the second single-line conveying device **152** can be transferred to neck handling (for example by arranging a rotating star and a locking device for successively feeding the containers to the rotating star). Following this, a transfer to the actual container treatment machines in the third treatment block **103** can take place.

[0062] The third treatment block **103** can further comprise a container exit point **132**, via which containers can, for example, be fed to further container treatment machines or further processed as packaged pallets. However, this is not essential for the invention.

[0063] If a fourth treatment block **104** is provided, it comprises a third container exit point **142**, via which containers can be fed to a third single-line conveying device **151**. Analogously to the second container exit point **122** of the second treatment block **102**, this can also cause containers from the mass conveyor **143** to be separated into singles via suitable gates before being transferred to the third single-line conveying device **151**.

[0064] If a fourth treatment block **104** is provided, the first treatment block **101** further comprises a third container feed point **111** via which containers from the third single-line conveying device **153** can be fed to the first treatment block **101** and in particular to a container treatment machine of this treatment block. Here, corresponding to the embodiments already described above, the containers can be suitably separated into singles or the containers can be spaced apart, for example by means of rotating stars and/or locking devices, in order to then feed them to the respective container processing machines.

[0065] The fourth treatment block **104** may further comprise a container feed point **141** via which containers can be fed to the fourth treatment block **104**. The container feed point can be designed analogously to the embodiments described so far.

[0066] The first treatment block **101** and/or the third treatment block **103** can also comprise a lane divider in the respective container feed point, which causes a division of the containers from the single-line transportation into a plurality of lanes in which the containers are however transported in an ordered manner and in a line one behind the other. From these lanes, the containers can then be fed to container processing machines. For example, each lane can lead to a different labeling machine. The throughput rate of treatment blocks **101** and **103** can thus be increased.

[0067] Furthermore, a control device **160** is shown in FIG. **1** which is connected to each of the provided treatment blocks **101**, **102**, **103** and **104**, preferably for bidirectional data exchange. In addition, the control unit **160** can also preferably be connected to the respective single-line conveying devices for bidirectional data exchange. The control unit can be designed as a control unit of the container treatment system and control all components of the container treatment system. Alternatively or additionally, each of the treatment blocks **101** to **104** can be assigned a respective control unit which can take over the control of the functions of the treatment blocks. The control units can in particular be designed as computers.

[0068] FIGS. **2a** and **2b** show different embodiments of treatment blocks arranged one after the other, wherein at least one of the treatment blocks has a mass conveyor. The first treatment block **101** and the second treatment block **102**, which comprises a mass conveyor, are therefore shown as examples in FIGS. **2a** and **2b**.

[0069] In FIG. **2a**, the transportation from the first treatment block **101** to the second treatment block **102** takes place via the single-line conveying device **152** in which the containers **180** are transported in a single line. At the first container feed point **121**, the containers from the single-line conveying device are then divided, as described with reference to FIG. **1**, for example via suitable gates, so that they can be fed to the mass conveyor **123** preferably over its entire width.

[0070] Here it is preferred if the mass conveyor **123** is moved more slowly than the transportation

speed of the containers **180** in the first single-line conveying device so that the capacity of the mass conveyor **123** is utilized as fully as possible. In particular, the transportation speed of the containers in the mass conveyor can correspond to the reciprocal value of the maximum number of transported containers across the width of the mass conveyor perpendicular to the transportation direction of the containers in the mass conveyor **123** multiplied by the transportation speed of the containers in the first single-line conveying device **152**. For example, if ten containers can be transported across the width of the mass conveyor perpendicular to the transportation direction of the containers in the mass conveyor, the transportation speed of the containers in the mass conveyor **123** can correspond to approximately one tenth of the transportation speed of the containers in the conveying device **152**. A lower belt occupancy in the conveying device of the mass conveyor is also conceivable.

[0071] Furthermore, the transportation speed of the containers in the mass conveyor can also correspond to the reciprocal value of the maximum possible average occupancy per line under the assumption of the densest packing of the bottles multiplied by the transportation speed of the containers in the first single-line conveyor **152**. Theoretically, this densest packing is hexagonal packing, wherein one line of bottles is offset from the adjacent line by half the diameter of the bottles perpendicular to the transportation direction. This tight packing is often not achieved perfectly, but only to a certain percentage. If, during operation, the mass conveyor can be occupied at, for example, 90% or 97.5% of the densest possible packing, this value can be used to determine the transportation speed in the single-line conveying device.

[0072] It can also be provided that the transportation speed of the containers in the mass conveyor **123** is used as a guide value since the transportation speed of the containers in the mass conveyor **123**, if this is implemented as part of a pasteurizer, for example, is substantially determined by the treatment time required for that pasteurization of the products or by the residence time of the containers in the pasteurizer.

[0073] FIG. **2b** shows an alternative embodiment to FIG. **2a** in which the transportation of the containers in the first single-line conveying device takes place via two transportation routes **251** and **252** arranged at a vertical distance from one another, which together form the single-line conveying device **152**. These transport containers **180** are on different levels so that the throughput rate of containers that can be conveyed with the second single-line conveying device **152** can be increased.

[0074] In this embodiment, it can be provided that the containers in the first treatment block **101** are treated only on one level. For example, a carousel can only contain one level of container receptacles. In order to distribute the containers onto the transportation routes **251** and **252**, suitable guide devices can be used to divide the containers. For example, it can be provided that the container treatment takes place in the first treatment block at the height of the transportation route **252**, and the containers **180** are transferred from this height to a lower height corresponding to the transportation route **251** and fed to this by a suitable vertical guide device. This can be done, for example, by suitable transport stars, wherein every second container can be fed to the second transportation route **251** and the other containers to the first transportation route **252**. For example, the transport star can have lowering elements that make individual lowering of the containers possible.

[0075] In the embodiment shown in FIG. **2b**, the second treatment block is designed such that the mass conveyor comprises two mass conveyors **231** and **232**, each at the level of the transportation route **251** and **252**, respectively.

[0076] The first container feed point **121** is then preferably designed such that it can implement a distribution of the containers transported in a single line in the respective transportation routes **251** and **252** to the respective mass conveyors **231** and **232**.

[0077] Alternatively to this, it can be provided that in the second treatment block **102** according to FIG. **2a**, the mass conveyor transports containers only on one level. In this embodiment, it can then

be provided that the first container feed point **121** enables transference of the containers, for example, from the transportation route **252** to the height of the mass conveyor **231**. This can be implemented analogously to the first container exit point **112** by suitable rotating stars which make possible a lowering of all containers arriving on the transportation route **252**. This can be designed in such a way that two independent gates are arranged at the level of, but upstream from, the mass conveyor **231**, which ensure distribution of the containers onto the mass flow **231** so that there are no accidental collisions between containers. The first arrangement of gates can distribute the containers of the transportation route **251** at a constant height. The second arrangement of gates can distribute the containers lowered from the transportation route **252** before they are fed to the mass conveyor **231** with the other containers.

[0078] At the second container exit point **122**, depending on the downstream single-line conveying device (not shown here), the containers can be arranged either on just one transport level or on two transport levels.

[0079] For example, the containers from the respective mass conveyors **231** and **232** can first be separated again into singles (via corresponding gates) and then transferred to the downstream single-line conveying device by being transferred to corresponding rotating stars with suitable guide devices for lowering or raising the containers.

[0080] FIG. **3** shows an example of a possible apparatus **300** for conveying and temporarily storing articles (not shown), in particular the containers described so far, which is configured to divide or separate a single-lane or single-track inlet stream **326** from a container feed point according to one of the previous embodiments with an exemplary possible conveying direction **340** into at least two article inlet paths **327**, **328**.

[0081] To divide or separate the inlet stream **326**, the apparatus **300** has an exemplary article inlet stream divider **323**, for example a switch, which can optionally guide or divide articles entering via the inlet stream **326** onto a (first) article inlet path **327** or onto another (second) article inlet path **328**.

[0082] For this purpose, the article inlet flow divider **323** can, for example, switch between two configurations or positions **324**, **325**. Shown here as an example, the article inlet stream divider **323** is in a first position/configuration **324** in which an article entering via the inlet stream **326** can be guided onto an article inlet path **328**.

[0083] The exemplary track or route or trajectory of an article in the configuration shown is represented by way of example using gray scale or hatching.

[0084] In addition, a possible second position/configuration **325** of the article inlet stream divider **323** is shown by way of example, in which an article entering via the inlet stream **326** would/can be directed to another (second) article inlet path **327**.

[0085] The apparatus **300** also has, for example, two movable article inlet distributors **321**, **322** which are in each case assigned to an article inlet path **327**, **328**. Here, for example, the article inlet distributor **321** is assigned to the article inlet path **327**, and the article inlet distributor **322** is assigned to the article inlet path **328**.

[0086] The article inlet distributors **321**, **322** can be designed, for example, as movable or displaceable slides or transfer slides, or carriages or transfer carriages.

[0087] The movable article inlet distributors **321**, **322** are configured to distribute articles from the article inlet paths **327**, **328** assigned to them to at least one intermediate storage apparatus **339**.

[0088] The movable article inlet distributors **321**, **322** are movable, displaceable or slidable, for example, along linear guides **331**, **332**, e.g. rails or grooves.

[0089] In addition, the article inlet distributors **321**, **322** can, for example, be designed in such a way that they can guide articles from one of the article inlet paths **327**, **328** to different regions of the intermediate storage apparatus **339**, for example to different rows of a multi-row buffer table or row buffer table, by means of non-rectilinear guide elements **333**, **334**. The possible guide elements **333**, **334** can, for example, be designed as guide walls or guard rails to guide the articles.

Preferably, this can be, for example, a conveyor channel with at least one belt, wherein the containers can be guided into the rows, for example, by a one or two-sided clamping.

[0090] The buffer apparatus **339** shown as an example is depicted as a multi-row buffer table or row buffer table, which can have a plurality of rows or tracks **301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315, 316, 317, 318, 319, 320** for receiving, conveying and temporarily storing articles.

[0091] For example, the rows or tracks **301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315, 316, 317, 318, 319, 320** of the buffer table or row buffer table are straight and parallel to each other. However, other geometries, e.g. non-rectilinear geometries, of the rows or tracks of the buffer table or row buffer table are also conceivable.

[0092] In the depicted exemplary configuration of the apparatus **300** or the configuration of the article inlet distributors **321, 322**, the article inlet distributor **321** is configured, for example, to direct articles picked up via the article inlet path **327** to the row **305**, while the article inlet distributor **322** is configured, for example, to direct articles picked up via the article inlet path **328** to the row **311**.

[0093] The article inlet distributors **321, 322** can, among other things, be configured to distribute articles to different, independent regions of the intermediate storage apparatus **339**.

[0094] For example, the article inlet distributor **321** can be configured to serve the rows or tracks **301, 302, 303, 304, 305, 306, 307, 308, 309, 310**, i.e. to distribute articles to the rows or tracks **301, 302, 303, 304, 305, 306, 307, 308, 309, 310**, while, for example, the article inlet distributor **322** can be configured so as to serve the rows or tracks **311, 312, 313, 314, 315, 316, 317, 318, 319, 320**, i.e. to distribute articles to the rows or tracks **311, 312, 313, 314, 315, 316, 317, 318, 319, 320**.

[0095] The article inlet distributors **321, 322** can be designed in such a way that they have at least partially complementary contours so that they can, for example, be displaced into one another in such a way that adjacent rows or tracks can also be served or approached by different article inlet distributors **321, 322**. For example, it is accordingly conceivable that line **310** can be served by article inlet distributor **321**, while article inlet distributor **322** is in a position in which it can/could serve line **311**.

[0096] The reference signs **329, 330** indicate exemplary possible sections of the article inlet paths **327, 328** for feeding articles into the intermediate storage apparatus. In other words, depending on the configuration or the position of the article inlet distributors **321, 322**, the different sections of the article inlet paths **327, 328** are usable or navigable, or can be used as a conveying path for articles.

[0097] For the sake of completeness, it should be mentioned that all sections of the shown paths, tracks and rows, i.e. inlet stream **326**, article inlet paths **327, 328, 329, 330**, as well as rows **301** to **320**, can be designed as conveyor paths that can be configured to transport articles such as containers.

[0098] As described above, drive means such as chain hoists and/or belts, as well as drives such as linear motors or long-stator linear motors can be used to drive the conveyor paths and/or to move the article inlet distributors **321, 322**.

[0099] Drive motors **335, 336, 337, 338** are shown by way of example, which can be configured together to drive the rows or tracks **301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315, 316, 317, 318, 319, 320** of the buffer apparatus.

[0100] In principle, any combination of the number and task of the drive motors **335, 336, 337, 338** as well as their coupling to the rows/tracks **301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315, 316, 317, 318, 319, 320** is conceivable.

[0101] It is conceivable, for example, that only two motors, e.g. **335** and **337**, are provided, wherein one of the motors drives the row/track into which containers are currently being conveyed or buffered, while for example the second motor drives the row or track that is currently being emptied or unbuffered.

[0102] It is also possible that, for example, two motors **335**, **336** can each drive at least one row/track of a group of rows/tracks, and at least one further motor **337** is/are provided for emptying or unbuffering at least one row/track.

[0103] It is also possible for each row/track to have its own drive.

[0104] Said exemplary drive motors can be designed, for example, as internal-rotor motors, which can be mounted on a shaft, for example, which can also be used to deflect a possible drive chain of a row/track.

[0105] As also mentioned, but not shown, the apparatus **300** can also have at least one or at least two movable article outlet distributors, wherein the article outlet distributors can be configured to receive articles from the at least one intermediate storage apparatus and to distribute them to at least one article outlet stream, in particular in the region of the container exit point according to one of the preceding embodiments and from there to the single-line conveying device.

[0106] FIGS. **4a** to **4c** show different designs of the container treatment system in order to illustrate an assortment of container treatment systems. In FIG. **4a**, the container treatment system **480** is arranged in a room **401** such as a factory hall which has an elongated shape. To take this into account, the respective treatment blocks **411** to **414** are arranged one after the other in a line in the transportation direction of the containers. The single-line conveying devices **415** to **417** extend between them and connect the container exit point of one treatment block to the downstream container feed point of a next treatment block.

[0107] Since buffering containers, if necessary, is already realized by the treatment blocks with a mass conveyor, corresponding buffer regions in the single-line conveying devices **415** to **417**, which would take up a lot of space, can be dispensed with. The single-line conveying devices **415** to **417** can be adapted in particular to the space requirements of the room **401**, in particular shortened or lengthened, depending on the available space. The treatment blocks **411** to **414** can, however, always be of identical design regardless of the length of the space **401**, so that an adaptation to a shorter space in the transportation direction of the containers can be achieved by shortening the single-line conveying device when planning the container treatment system **480**. However, a structural redesign of the treatment blocks is unnecessary, which facilitates the construction of the container treatment system **480**.

[0108] It can also be provided that the treatment blocks are always of identical design except for the arrangement of the container feed points and/or exit points, so that by adjusting the single-line conveying devices and the arrangement of the container feed points and exit points, adaptation to spatial conditions is achieved without the treatment blocks themselves having to be changed.

[0109] FIG. **4b** shows another design of the container treatment system **480**. Here, the available space **402** is basically rectangular and does not permit, for example, a linear arrangement of the treatment blocks **421** to **424** one after the other, as was the case in FIG. **4a**. Since the treatment blocks are preferably basically of the same design as the treatment blocks **411** to **414** corresponding to FIG. **4a**, an adaptation to the spatial conditions can be made by changing the single-line conveying devices **425** to **427**, wherein, for example, the single-line conveying device **426** realizes a transportation of the containers along a curve so that the conveying device of the containers is reversed and space can be saved. Depending on the length and width of the room **402** (e.g. a factory hall), the length and/or curvature of the individual single-line conveying devices can be adjusted if necessary.

[0110] As an alternative to the embodiments in FIGS. **4a** and **4b**, FIG. **4c** shows a container treatment system **480** which is to be arranged in a substantially L-shaped space **403**. Here too, treatment blocks **431** to **434** are provided which are of substantially identical design to the treatment blocks according to the embodiments in FIGS. **4a** and **4b**. An adaptation to the available space **403** can here again be achieved by suitable designs of the single-line conveying devices **435** to **437**, whereby here in the conveying device **436** the transportation direction is changed by 90° in order to take into account the L-shape of the available space **403**.

[0111] Since the single-line conveying devices do not have to ensure a buffering effect, they can be selected practically as desired in terms of their length and therefore their container capacity (the maximum number of containers present in the respective single-line conveying device at a given time) in order to enable adaptation to the spatial conditions. The buffering effect is ensured by the treatment blocks themselves, which are basically always of identical design.

[0112] With the embodiments presented as examples in FIGS. 4a to 4c, an assortment of container treatment systems can be realized that can be adapted practically as desired to the available space, for example in a factory hall. This is achieved by adapting the single-line conveying devices for single-line transportation between the treatment blocks, whereas the individual treatment blocks are preferably always of identical design. For container treatment systems in the assortment, minor differences in the treatment blocks can be provided, in particular with regard to the container feed point and/or the container exit point. This concerns in particular transportation on one or more transportation levels but also, for example, the exit angle of the containers from the respective container exit point or the entry angle into the respective container feed point. This depends on the orientation of the conveying devices and can be suitably modified.

[0113] However, the container treatment machines within the respective treatment blocks are preferably always of identical design and construction for each container treatment system of such an assortment.

Claims

1. A container treatment system according to the invention for treating containers, comprising a first treatment block with a first single-line container exit point, a second treatment block arranged downstream comprising a wet region and/or temperature control region with a mass conveyor for containers or a buffer system for containers, a first single-line container feed point upstream from the mass conveyor or the buffer system, and a second single-line container exit point downstream from the mass conveyor or the buffer system, and a third treatment block arranged downstream from the second treatment block with a second single-line container feed point, wherein the single-line container exit point of the first treatment block is connected to the single-line container feed point of the second treatment block by a first single-line conveying device, and the single-line container exit point of the second treatment block is connected to the single-line container feed point of the third treatment block by a second single-line conveying device.
2. The container treatment system according to claim 1, wherein the wet region and/or temperature control region is designed as a pasteurizer, and/or the second treatment block comprises a blower for blowing off liquid on the surface of the containers; or wherein the buffer system comprises a row buffer comprising at least two single-line buffer tracks, or a mass buffer for the unordered buffering of containers.
3. The container treatment system according to claim 1, wherein the first single-line container exit point, the first single-line conveying device and the first single-line container feed point are designed for the single-line transportation of containers on at least two vertical transportation routes; and/or wherein the second single-line container exit point, the second conveying device and the second single-line container feed point are designed for the single-line transportation of containers on at least two vertical transportation routes.
4. The container treatment system according to claim 1, wherein the first treatment block comprises a filler, and/or a capper, and/or an inspection device for inspecting the containers.
5. The container treatment system according to claim 1, wherein the third treatment block downstream from the second container feed point comprises a lane divider for dividing the containers from the second single-line conveying device into a plurality of lanes, and/or wherein the third treatment block comprises an inspection device for inspecting the containers, and/or wherein the third treatment block comprises a labeling machine and/or a direct printing machine.

6. The container treatment system according to claim 1, wherein the second treatment block and/or the third treatment block comprises a coding device for applying a code to containers.
 7. The container treatment system according to claim 1, wherein no treatment of containers takes place in the region of the first conveying device and/or in the region of the second conveying device.
 8. The container treatment system according to claim 1, wherein upstream from the first treatment block, a fourth treatment block is arranged with a wet region and/or temperature control region, comprising a mass conveyor for containers, or a buffer system for containers, wherein the fourth treatment block comprises a third single-line container exit point, and the first treatment block comprises a third container feed point, wherein the third single-line container exit point is connected to the third single-line container feed point by a third single-line conveying device.
 9. The container treatment system according to claim 8, wherein the fourth treatment block comprises a container washing machine.
 10. The container treatment system according to claim 8, wherein the third single-line container exit point, the third conveying device and the third single-line container feed point are designed for the single-line transportation of containers on at least two vertical transportation routes.
 11. The container treatment system according to claim 8, wherein the fourth treatment block and/or the second treatment block comprise a buffer region for the random buffering of containers outside the wet region and/or temperature control region and downstream from the container feed point of the treatment block and upstream from the container exit point of the treatment block.
 12. An assortment of container treatment systems, wherein each of the container treatment systems comprises a container treatment system according to claim 8, and wherein for two different container treatment systems of the assortment of container treatment systems, the first single-line conveying device, and/or the second single-line conveying device, and/or the third single-line conveying device have different forms.
 13. The assortment of container treatment systems according to claim 12, wherein for two different container treatment systems in the assortment of container treatment systems, the treatment blocks are substantially of identical design.
 14. A method for treating containers is provided, wherein the containers are treated in a container treatment system according to claim 1, and at least one treatment step of the containers is carried out in the first treatment block, the second treatment block and the third treatment block.
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